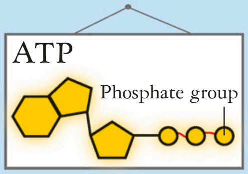
CONTROL CENTER

The NUCLEUS is found only in eukaryotic cells. It acts as a control center, sending and receiving messages to and from other cells, and deciding how the cell works. It also controls how the cell grows and replicates itself. The blueprint for the cell, its DNA, is held in the nucleus. The DNA is coiled into strands, called chromosomes. The region of a chromosome that tells the cell how to make a specific protein is called a gene. Human cells contain around 30,000 genes arranged on 46 chromosomes. When the cell needs to make new proteins part of the DNA unzips and copies the right sequence for that protein.

ATP MOLECULE STORE

EXIT TO REST OF CELL



ATP is vital to the working of a cell. It helps move substances in and out of the cell membranes, supplies the energy needed for work, and acts as a switch to control chemical reactions. Its energy is released by splitting off one or more of its phosphate groups. Each cell contains around one billion molecules of ATP, which are constantly being used and recycled.

We've split nearly a billion molecules today.

Phosphate groups

CELL MEMBRANE

LIFT TO ATP STORE

TO MITOCHONDRIA

In the mitochondria the molecules undergo a second set of reactions called the citric acid cycle that turn the pyruvate into carbon dioxide and water and create more molecules of ATP.

ATP for storing

ATP

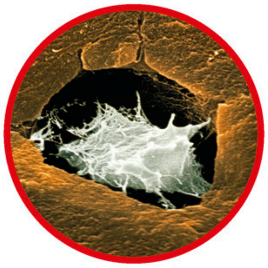
Pyruvate for processing

SPECIAL CELLS

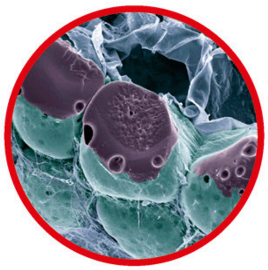
The human body contains around 200 different types of cells. Some group together to form the tissues that make up the body's organs, such as the brain, heart, skin, and lungs. Within these tissues are highly specialist cells that do a particular job. There are blood cells, hair cells, fat cells, bone cells, pigment cells, and eye cells that help you see in color and black and white.

BLOOD CELLS are used to carry oxygen around the body and collect carbon dioxide. They are replaced every 120 days.

NEURONS are nerve cells that transmit signals to and from all parts of the body. Some can be several feet in length.



BONE CELLS are made in the marrow of the large bones. As they develop and grow they harden to give the bone strength.



FAT CELLS are used to store energy and insulate the body against heat loss. They are mainly found under the skin and around the major organs.

1 billion cells every hour.